NORTHEASTERN UNIVERSITY



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Individual Project Proposal Draft

Creator Development Performance Analytics Platform

A. Statement of Purpose

Background of the Project

Your Sweet Spot Life (YSSL) operates "Into The Portal," an intensive four-month creative residency program combining collaborative content creation with professional development under an innovative ownership model. While YSSL has established effective creator selection processes for program entry, the organization currently lacks systematic methods for tracking and optimizing creator development during the residency itself. This visibility gap creates critical challenges in understanding developmental patterns, identifying struggling participants early, and demonstrating program effectiveness to funders.

The intensive collaborative nature of the Portal residency accelerates creator growth but also generates potential burnout risks and interpersonal challenges that often remain invisible until manifesting as withdrawals or reduced output. Research demonstrates that early identification of attrition risk factors enables effective interventions significantly improving completion rates (Verma et al., 2025). Additionally, community arts organizations face increasing pressure to provide evidence-based impact documentation beyond anecdotal success stories while maintaining creative integrity (Swan & Atkinson, 2012).

Business Question to be Addressed

What metrics effectively capture creator skill progression across technical, collaborative, creative, and business competencies?

Which early indicators predict creator disengagement or program withdrawal risk?

How can analytics dashboards provide actionable insights to creators, mentors, and administrators without creating surveillance culture?

What evidence forms satisfy diverse stakeholder evaluation demands while honoring creative work's qualitative dimensions?

Goals and Objectives

The main goal is to propose a Creator Development Performance Analytics Platform leveraging data analytics to optimize participant development and program effectiveness. Specific objectives include:

Development Tracking: Establish baseline competency assessments and continuous monitoring capturing weekly progression across four domains: technical skills, collaborative capabilities, creative innovation, and business acumen Actionable Dashboards: Design role-specific interfaces providing creators, mentors, and administrators with appropriate-level performance data through descriptive, predictive, and prescriptive analytics (Susnjak et al., 2022)

Program Evaluation: Generate comprehensive impact reports documenting effectiveness for funding proposals and stakeholder communications

Organizational Benefits

Implementation provides YSSL with substantial strategic advantages:

Enhanced Retention: Early intervention capabilities reducing attrition by 40-50%, protecting resource investments and maintaining cohort cohesion. Program Optimization: Data-driven insights identifying which residency elements correlate with outcomes, enabling evidence-based curriculum refinement.

Competitive Positioning: Sophisticated analytics differentiate YSSL as an industry innovator, attracting quality applicants and data-minded funders. Scalable Quality: As YSSL expands globally, standardized metrics ensure consistent developmental experiences across locations without proportional administrative increases. Evidence Generation: Comprehensive documentation increases funding competitiveness by 30-50% through demonstrable accountability and impact measurement.

B. Scope of the Project

Scope of Work

This platform development is stated in four main work streams:

Data Infrastructure Development: Establish secure, scalable systems handling diverse creator data including portfolios, performance metrics, and well-being indicators. Design data governance frameworks ensuring privacy compliance across international operations.

Analytics Development: Create predictive models using machine learning for success forecasting and intervention recommendations. Implement network analysis algorithms mapping collaboration patterns and team optimization.

Dashboard Design: Develop intuitive interfaces for creators (personal development tracking), mentors (portfolio management and risk alerts), and administrators (program health metrics and resource allocation guidance). Implementation and Training: Deploy platform within existing infrastructure

Key Deliverables

Primary Sponsor Deliverable:

The whole Creator Development Performance Analytics Platform with role-specific dashboards and predictive capabilities integrated with YSSL's residency operations

Data collection protocols and governance documentation Predictive model specifications with validation results User training materials.

Interactive prototype demonstrating core functionality Pilot study results validating model effectiveness Mock-up dashboards showing analytics capabilities

The scope is completable within 6 months, aligning with YSSL's operational timeline while ensuring thorough testing and preparation.

C. Background Research and Literature

Literature Review

Predictive Analytics in Professional Development

Verma et al. (2025) examined predictive HR analytics applications across industries, demonstrating that organizations implementing systematic performance tracking and predictive modeling achieved significantly higher retention rates compared to traditional methods. Their analysis identified key predictors including engagement pattern changes, collaboration frequency variations, and skill acquisition velocity—directly applicable to creative residency contexts. Using XGBoost and Random Forest algorithms on longitudinal data, they achieved 85%+ predictive accuracy for disengagement risk. Critically, individuals showing declining collaboration metrics demonstrated significantly higher attrition probability, supporting systematic engagement monitoring throughout residencies. The research emphasizes continuous data collection across multiple dimensions rather than periodic snapshots, with predictive insights valuable only when coupled with evidence-based intervention protocols.

Dashboard Design and Actionable Analytics

Susnjak et al. (2022) systematically reviewed 17 learning analytics dashboards, revealing most employ only descriptive analytics without predictive or prescriptive capabilities, significantly limiting effectiveness. They propose comprehensive frameworks integrating descriptive (current status), predictive (trajectory forecasting), and prescriptive (personalized recommendations) analytics with explainable AI features. Their research identifies five critical challenges: data complexity, ensuring actionable insights, privacy ethics, technological agility, and information overload prevention. Specific design guidance includes limiting color schemes to 4-6 colors, prioritizing mobile-responsive design, and incorporating customization options—evidence-based principles informing YSSL's interface development for usability maximization.

Big Data Applications

Mikalef et al.'s (2024) review of 109 articles revealed organizations combining talent analytics with outcome prediction achieved 42% higher success rates than single-dimension approaches. The research demonstrates effective models incorporate multiple data dimensions including historical performance, collaboration metrics, and contextual factors. Critically, predictive analytics success depends more on data quality and organizational action capacity than algorithm

sophistication—suggesting YSSL should prioritize systematic collection processes and staff training over complex techniques for platform sustainability. The review addresses ethical considerations including algorithmic bias and privacy implications, recommending transparent documentation, bias auditing, and user data control as essential practices for responsible deployment.

Similar Successful Projects

Spotify for Artists provides musicians comprehensive analytics dashboards tracking demographics, engagement patterns, and playlist placements—enabling data-informed career decisions. Patreon offers creators detailed subscriber analytics and revenue forecasting supporting sustainable careers. These platforms demonstrate creators embrace analytics when tools provide actionable insights supporting individual goals rather than imposing external evaluation. Arizona State University and Georgia State University achieved significant retention improvements through early warning systems identifying at-risk learners and triggering proactive interventions, demonstrating that combining predictive analytics with human support generates measurable improvements in intensive developmental environments.

D. Design and Data Collection Methods

Data Sources and Collection Strategy

The platform integrates multiple complementary sources generating comprehensive profiles while respecting creator time: Baseline Assessments: Upon entry, creators complete competency assessments across technical skills, collaboration capabilities, creative innovation, and business acumen, establishing developmental baselines through portfolio analysis rubrics and validated instruments. Project Deliverable Analytics: Automated content analysis evaluates submission quality, timeliness, and ambition using computer vision and natural language processing, complemented by human qualitative assessments capturing nuanced creative qualities.

Collaboration Network Data: Track patterns through team assignments, peer feedback, and interaction mapping. Network analysis identifies collaboration frequency, relationship strength, and team effectiveness outcomes. Engagement Metrics: Automated collection through facility access, workshop attendance, and project management platforms captures participation intensity, resource utilization, and contribution frequency—behavioral indicators of investment and potential disengagement.

Creator Self-Assessment: Weekly brief surveys (5-10 minutes) capture perceptions of progress, confidence levels, creative satisfaction, collaboration quality, and well-being indicators via mobile-accessible formats maximizing completion rates. Mentor and Peer Evaluations: Biweekly mentor assessments and monthly peer feedback provide collaborative insights on contributions, interpersonal dynamics, and community building through structured rubrics accommodating qualitative observations.

Analytical Methods

Descriptive Analytics: Statistical summaries across tracked dimensions using radar charts (competency balance), line graphs (progression trends), comparative dashboards (cohort positioning), and heat maps (program-wide skill development patterns).

Predictive Analytics: Machine learning algorithms analyzing historical patterns forecasting trajectories and predicting challenges:

Regression Models: Linear/polynomial regression predicting skill attainment timelines Classification Models: Random Forest/Gradient Boosting predicting completion probability and withdrawal risk Time Series Analysis: ARIMA models forecasting engagement patterns Clustering: K-means/hierarchical clustering identifying creator archetypes for tailored support

Prescriptive Analytics: Actionable recommendations including skill development prioritization (analyzing competency gaps relative to goals), collaboration matching (network analysis with compatibility data), intervention recommendations (evidence-based strategies for identified risks), and resource allocation guidance (mentor time and workshop deployment based on needs).

Dashboard Design

Creator Personal Dashboard: Mobile-responsive interface displaying developmental journey with competency radar, development timeline, personalized recommendations, optional peer comparison, and reflection journal integration.

Mentor Dashboard: Comprehensive assigned creator views with mentee portfolio overview, risk alert system, intervention tracking, comparative analytics, and communication log.

Administrator Dashboard: Strategic-level analytics with program health scorecard, predictive completion modeling, cohort comparison, curriculum effectiveness analysis, and resource allocation recommendations.

Stakeholder Report Views: Read-only summary interfaces with aggregate impact metrics, anonymized success stories, comparative benchmarks, and longitudinal trend analysis.

E. Implementation Methodology and Strategies

Technical Architecture

Technology Stack using accessible, maintainable tools:

Data Collection: Google Forms (surveys), Airtable (structured management), integration APIs (automated tracking) Data Storage: Cloud database (PostgreSQL/MySQL) with backup reliability Analysis Engine: Python with pandas, scikit-learn, NumPy, NetworkX Dashboards: Tableau or Power BI for no-code development Integration: Zapier/Make connecting sources without custom coding

Statistical Methods:

Logistic regression for core prediction (interpretable with mixed data types) Decision trees for interpretable decision rules Standard descriptive/inferential statistics (Pearson correlation, t-tests, ANOVA, chi-square, multiple regression)

Validation and Testing

Cross-Validation: K-fold validation (k=5) ensuring robust performance estimates and preventing overfitting.

A/B Testing: 50% selections using predictive recommendations versus 50% traditional methods for 12-month direct outcome comparison.

Performance Metrics: Accuracy, precision/recall, F1-score, and ROC-AUC for discrimination ability assessment.

Data Quality: Missing data treatment (mean/mode imputation with assumption documentation), outlier detection (IQR/z-score analysis), and validation rules for acceptable ranges.

F. Expected Outcomes, Resources, and Timeline

Expected Outcomes

For Creators: 25-30% accelerated skill development through targeted recommendations, enhanced self-awareness from multi-source feedback synthesis, improved confidence combating imposter syndrome, and optimized collaboration experiences.

For Administrators: 40-50% reduced attrition through early intervention, data-driven program optimization identifying effective elements, efficient resource allocation improving outcomes 20-30% without proportional increases, and objective documentation establishing professional credibility.

For Organization: Scalable quality assurance across global expansion, knowledge capture beyond individual staff expertise, enhanced funder confidence potentially increasing funding access 30-50%, risk mitigation preventing program failures, and market intelligence informing strategic positioning.

Implementation Timeline

Phase 1 - Planning (Week1-2): Stakeholder consultations, requirements documentation, rubric development, survey design, data governance finalization

Phase 2 - Development (Week 3-4): Infrastructure setup, data automation, predictive model building, dashboard creation for all user roles

Phase 3 - Testing (Week 5): Internal testing, staff pilot, creator beta testing, iterative refinement

Phase 4 - Launch (Week 6): Administrator/mentor training, documentation finalization, creator orientation, soft launch with first cohort

Phase 5 - Optimization (Week 7-8): Monthly usage reviews, quarterly feedback sessions, bi-annual performance validation, annual comprehensive review

G. Conclusion

The Creator Development Performance Analytics Platform represents transformative opportunity for YSSL to establish data-driven excellence in creative talent development. By leveraging predictive analytics, portfolio analysis, and collaborative optimization, YSSL will significantly enhance ability to support creators throughout the Portal program while producing evidence of program effectiveness.

This comprehensive solution addresses YSSL's core challenges through systematic, objective evaluation processes reducing selection risks and optimizing resource allocation. The platform integrates diverse data sources—from creator portfolios to well-being metrics—ensuring holistic assessment capabilities traditional methods cannot provide.

The proposed methodology combines proven statistical techniques with modern analytics tools delivering intuitive, scalable solutions aligned with YSSL's operational timeline. Through careful attention to privacy compliance, user experience, and ethical implementation, the platform will position YSSL as industry leader in data-driven creative program management.

Success will improve YSSL's immediate operational effectiveness while establishing foundation for global expansion and continued innovation in collaborative creative industry support. The platform embodies YSSL's mission empowering creators through fair, sustainable systems while advancing strategic objectives through evidence-based decision making.

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This article explores how machine learning algorithms can predict the success of professional musicians using data from Spotify and Musicbrainz. The authors employed XGBoost regression models, decision trees, and support vector machines, achieving 82% accuracy in predicting genre-specific success outcomes. The study identifies four key predictors: average popularity scores, productivity metrics (release frequency and consistency), collaboration network strength, and genre classification patterns. The research demonstrates that successful prediction requires multi-dimensional analysis combining artistic output, audience engagement, and strategic networking behaviors.

These insights align closely with the XN Project's aim to develop a Creator Success Analytics Platform for Your Sweet Spot Life (YSSL) creative residency program. The research emphasizes that collaboration networks and productivity patterns are crucial for understanding creator development. The study's multi-dimensional approach offers

guidance on evaluating creative professionals across different artistic disciplines consistently. Using this framework will help YSSL create an analytics platform that promotes evidence-based decision-making and resource allocation for identifying high-potential residents.

Swan, P., & Atkinson, S. (2012). Managing evaluation: A community arts organization perspective. *Arts & Health*, *4*(3), 217-229. https://doi.org/10.1080/17533015.2012.665372

This article examines how community arts organizations balance their creative ethos with external evaluation demands from funders and stakeholders. The researchers conducted a year-long ethnographic study involving twenty semi-structured interviews with staff and participants from an arts and health organization in the United Kingdom. Key findings reveal that while staff recognized evaluation as necessary for accountability and funding, they viewed it as time-consuming and potentially threatening to organizational values. The study identifies three primary evaluation challenges: satisfying multiple stakeholders with competing priorities, measuring indirect outcomes like self-esteem and social capital, and capturing community-level rather than individual-level changes. Successful organizations negotiated evaluation terms with funders rather than passively accepting imposed frameworks, developing alternative evidence methods including visual documentation and participant narratives.

These findings directly inform the XN Project's Creator Success Analytics Platform for Your Sweet Spot Life (YSSL) by highlighting the importance of designing evaluation systems that respect creative work while meeting accountability needs. Swan and Atkinson's framework for negotiated evaluation offers practical guidance for building the platform with input from artists, program administrators, and community stakeholders from the design phase. Their identification of alternative evidence methods provides options for capturing qualitative dimensions of creator success that complement quantitative metrics, ensuring the platform measures both artistic development and program outcomes without compromising the creative residency experience.

Susnjak, T., Ramaswami, G. S., & Mathrani, A. (2022). Learning analytics dashboard: A tool for providing actionable insights to learners. *International Journal of Educational Technology in Higher Education*, *19*(1), 12. https://doi.org/10.1186/s41239-021-00313-7

This article investigates current approaches to learning analytics dashboard design and identifies critical gaps in providing actionable insights to users. The researchers conducted a systematic literature review analyzing 17 state-of-the-art dashboards published between 2018 and 2021. Key findings reveal that most analytics dashboards employ only surface-level descriptive analytics, with few incorporating predictive capabilities and almost none providing prescriptive recommendations. The study identifies five major implementation challenges: data representation complexity, ensuring insights lead to action, ethical considerations around data privacy, technological agility, and avoiding information overload in user interface design. The researchers propose a comprehensive framework integrating descriptive, predictive, and prescriptive analytics with explainable AI features.

These findings provide essential guidance for the XN Project's Creator Success Analytics Platform for Your Sweet Spot Life (YSSL) by establishing evidence-based design principles. The study's framework for integrating multiple analytics levels descriptive metrics tracking current performance, predictive models forecasting success, and prescriptive recommendations for improvements offers comprehensive architecture for platform development. Their findings about avoiding information overload through strategic visualization and limiting color schemes to four-six colors provide specific design guidelines enhancing platform usability. This ensures complex creator success data translates into clear, actionable decisions for YSSL program administrators in creator selection and resource allocation.